42 BERLIN × NEEDLE · AGENT HACKATHON

NEEDLE AGENT

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79/150
PUBLIC · 52.7%

31/100
HIDDEN · 31%

1187
COMMITS SINCE 19:45

A self-restarting, watchdog-guarded, multi-tier coding agent that overnight ground a knitting compiler out of nothing.

the rules of the game

- Read a hidden spec at 20:00 Thursday
- Build the program it describes
- Submit by 12:00 Friday
- Use only **free** or **local** models — no paid Claude/GPT/Copilot
- Judges grade: 40% hidden tests + 20% public + 20% process + 10% self-tests + 10% talk
- The agent is the project. The program is just the battlefield.

architecture — what we built

MODEL ROUTER

Cerebras Cloud (Qwen3-235B) → Ollama local (Qwen3-Coder-30B) failover. 60 s 429 cooldown, retry+backoff, hard timeout.

QUERY LOOP

Per-iter: `read` → model → parse tool calls → execute → tool_result → repeat. Auto-commit per iteration.

9 TOOLS

read · write · edit · bash · glob · grep · run_tests · load_skill · spawn_subagent · submit_done. `edit` has read-before-edit + exact-match guards from Claude Code.

12 SKILLS

On-demand markdown injected via `load_skill`. Generic + 2 knitting-domain (error-format, stitch-math).

SAFETY NETS

cycle detector (3+ same call) · auto-compaction (>60 KB) · XML tool-call recovery · thinking-without-action nudge.

WATCHDOG

External bash loop: restarts agent on death · restores best `knit.py` on regression · pushes to GitHub every 10 min.

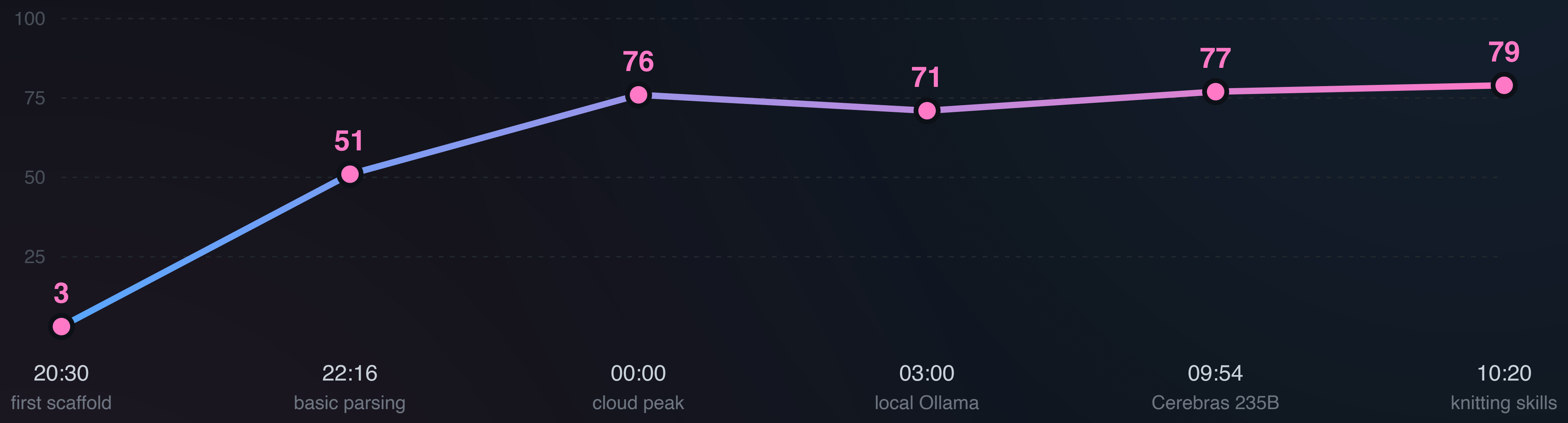
the reveal (20:00)

`secret_spec/SECRET_SPEC.md` — **Knitting Compiler**, 927 lines

```
python3 solution/knit.py compile <file.knit>
```

Parses a knitting DSL, validates, expands repeats, simulates stitch counts row by row, emits one deterministic JSON. 150 public tests across 8 levels of difficulty.

the journey — score over time



From 3 to 79/150 in 14 hours. Every dot is a `run_tests` result. The dip at 03:00 is the swap from cloud (regressed) to local Ollama; the climb after is Cerebras+local hybrid with knitting-domain skills.

the model carousel

TRIED	VERDICT
DeepSeek V4-Pro (Featherless)	`tools=false` — useless
DeepSeek V3.2 (Featherless)	empty-args bug
DeepSeek V3-0324 (Featherless)	markdown instead of native tools
Qwen3-Coder-480B (Featherless)	slow + drops
Qwen3-Coder-480B (NVIDIA NIM)	rate-limited fast
Qwen3-235B (**Cerebras free**)	~2000 tok/s — primary
Qwen3-Coder-30B UD-Q4_K_XL (local Ollama)	~80 tok/s — fallback
Qwen3-Coder-30B IQ4_NL (local)	broken — invented tool names like `_write`, args like `"Eight"`

the overnight grind — in numbers

906	AGENT DECISIONS
1187	GIT COMMITS SINCE CHECKPOINT
23	WATCHDOG AUTO-RESTARTS
808	CEREBRAS 429 COOLDOWNS
621	THINKING-WITHOUT-ACTION NUDGES
199	CONTEXT AUTO-COMPACTIONS
66	PROVIDER AUTO-FALLBACKS
42	RUN_TESTS INVOCATIONS
4	ALL-PROVIDERS-DOWN EVENTS

what worked

- **Per-iteration auto-commit** — `git log` is a play-by-play of the agent's thinking
- **Claude Code's edit invariants** — read-before-edit + exact-match killed an entire class of silent overwrites
- **Cycle detector** — 3+ identical tool calls in a 5-window → context prune
- **Auto-compaction** — old tool results squashed to head+tail when context > 60 KB
- **3-tier model failover** + 429 cooldown — kept the loop alive when Cerebras hit TPM
- **Watchdog with regression restore** — pulled the best `knit.py` back from commit history when the model overwrote it with garbage

hidden tests — post-mortem (31/100)

HIDDEN CATEGORY	SCORE	DIAGNOSIS
boundary_stress_valid	**14/15**	large patterns work, math holds at scale
cli_behavior	**4/5**	CLI contract solid
schema_exactness_valid_outputs	7/15	half — JSON field shape mismatches
compositional_valid_patterns	6/30	**systematic off-by-one in stitch math** — 24/24 failures show `expected N, got N+1`
edge_policy	**0/10**	not detecting invalid → exit 0 when judge expects 1
interaction_effect_invalid_patterns	**0/25**	same root cause as public level_05/06 — error emission shape never landed

Root causes (identifiable, not random):

- One stitch op (likely `yo` / `m1` / `kfb`) produces +1 too many stitches → 24 free points
- Whole error-emission pipeline never converged on the canonical `{type, code, line, row}` schema → 35 points trapped
- One more agent grind window aimed at level_05/06 would have closed both gaps

what failed

- **Free-tier rate limits** stalled us 30 min at a time when all 3 cloud providers were 429
- **IQ4_NL quantization** of Qwen3-Coder produced hallucinated tool calls — switched to **UD-Q4_K_XL**
- **Ollama registry** had EOF-on-manifest issues for `qwen3-coder:30b` — bypassed via `hf.co/unsloth/...:UD-Q4_K_XL`
- `submit_done` **premature calls** at sub-150 scores → tightened prompt to require ≥ 140
- **Level_05 (single errors) only 2/30** — knitting error JSON format was the biggest unsolved bucket

what we'd improve given more time

- A `verify_score` **tool** that wraps `run_tests` and lets the loop refuse `submit_done` when score < 140
- **High-water-mark snapshots inside the agent** (today we did it via an external watchdog)
- **Bigger local quant** (Q5_K_M / Q6_K) on hardware with more RAM
- **Spec-derived self-tests** generated proactively, not just at the end
- **Per-level subagents** that own one category and don't context-pollute the rest
- **Hidden-test post-mortem loop:** if we'd had a second 4-hour window we'd point a fresh agent at the off-by-one stitch math (worth +24 hidden points) and the canonical error schema (worth +35 hidden points). The failures are systematic — not noise — and the agent is already wired to grind specific bugs.

the agent is the project

```
agent-readiness-1945  ← Thursday 19:45 checkpoint (set before the spec was even open)
              └─ 1187 commits ─→ HEAD  (every iteration a commit, every commit a decision)
```

Public score is the battlefield score.

The agent is what we built.

https://github.com/HackatonWinnners/Agent007

thank you 🧶